

Machine-learning-based model predictive control of a solid oxide fuel cell with dusty-gas porous-anode transport

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Abstract. A dynamic first-principles model of a planar solid oxide fuel cell (SOFC), in which binary $\text{H}_2/\text{H}_2\text{O}$ transport through the porous anode is described by the dusty-gas model (DGM), is used to generate simulation data for a gated-recurrent-unit (GRU) surrogate. The surrogate is embedded in a model predictive controller (MPC) that tracks a power demand while respecting a cell-temperature limit, a minimum H_2 fraction at the triple-phase boundary (TPB) and a fuel-utilisation limit. The steady operating map shows that at high fuel utilisation the safe power is limited by H_2 depletion at the TPB, a porous-transport effect. In closed loop on the first-principles plant, the GRU-based MPC reduces the power-tracking error roughly four-fold relative to PI control with a fixed utilisation ratio and removes all constraint violations, at the cost of lower electrical efficiency.

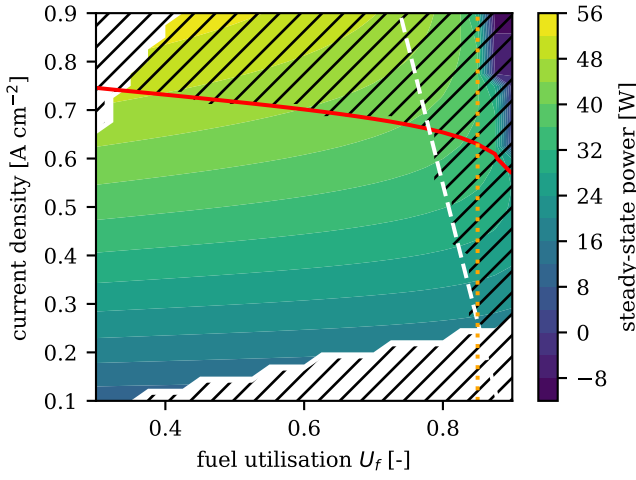


Figure 1: Steady operating map. Red: $T = 1100\text{ K}$; white dashed: $x_{\text{TPB}} = 0.10$ (DGM); orange: $U_f = 0.85$; hatched: unsafe region.

1. Introduction

SOFCs operating in load-following mode must change power quickly while avoiding thermal stress and fuel starvation of the anode. In anode-supported cells, gas transport through the porous electrode makes the composition at the TPB differ markedly from that in the fuel channel; the DGM, which includes Knudsen, molecular and viscous contributions, is a reliable description of this effect [1, 2, 3]. Detailed models of this kind are, however, expensive to embed in a real-time optimiser. Recurrent neural networks trained on first-principles simulations have been shown to be effective predictors for MPC of nonlinear processes [4, 5]. This work combines the two: a DGM-resolved TPB constraint is imposed through a GRU surrogate inside an MPC, and the resulting controller is compared with a conventional controller and with the same MPC using the first-principles model.

2. First-principles model

The anode (thickness $L = 1\text{ mm}$, porosity $\varepsilon = 0.3$, tortuosity $\tau = 5$, pore radius $r = 0.5\text{ }\mu\text{m}$) is treated as quasi-steady. For species i in the binary $\text{H}_2/\text{H}_2\text{O}$ mixture the DGM reads

$$\frac{N_i}{D_{iK}^e} + \frac{x_j N_i - x_i N_j}{D_{ij}^e} = -\frac{1}{RT} \frac{dp_i}{dz} - \frac{B_0 p_i}{\mu RT D_{iK}^e} \frac{dp}{dz}, \quad (1)$$

with $D^e = (\varepsilon/\tau)D$, Knudsen diffusivity $D_{iK} = \frac{2}{3}r\sqrt{8RT/\pi M_i}$, permeability $B_0 = \varepsilon r^2/8\tau$ and fluxes $N_{\text{H}_2} = -N_{\text{H}_2\text{O}} = i/2F$. Integrating from the channel to the

TPB gives x_{TPB} and p_{TPB} , and the cell voltage is

$$V = E_N(x_{\text{TPB}}, p_{\text{TPB}}, T) - \eta_a - \eta_c - i \text{ASR}(T), \quad (2)$$

with Butler–Volmer (asinh) activation losses and an Arrhenius ohmic resistance. The dynamics are a mole balance on a well-mixed anode compartment and a lumped energy balance,

$$N_V \dot{x}_{ch} = F_{in}(x_{in} - x_{ch}) - iA/2F, \quad (3)$$

$$M_{C_p} \dot{T} = iA(E_{in} - V) - hA(T - T_{amb}). \quad (4)$$

The manipulated inputs are the current density i and the fuel feed F_{in} ; the outputs are x_{ch} , T , V and x_{TPB} , sampled every 5 s.

3. Surrogate model and controller

Data. 2000 training, 300 validation and 300 test trajectories of 80 steps were simulated with amplitude-modulated pseudo-random binary (APRBS) inputs [6], with random hold times of 2–8 samples and random fuel utilisation between 0.30 and 0.92.

Surrogate. A small network maps the measured state to the initial hidden state of a GRU [7] (64 units), which is driven by the input sequence; an output layer returns the four outputs at every step (21.8k parameters). It was trained on 20-step windows with Adam for 60 epochs.

MPC. Over a 100 s horizon (20 steps, 5 move blocks) the controller minimises

$$J = \sum_k \left[(P_k - P_{ref})^2 + w_f F_{in,k} + w_\Delta \|\Delta u_k\|^2 + \rho \phi_k \right],$$

where ϕ_k penalises violations of $T \leq 1100\text{ K}$, $x_{\text{TPB}} \geq 0.10$ and $U_f \leq 0.85$ (soft constraints). The problem is solved with L-BFGS-B using gradients obtained by automatic differentiation through the GRU; the first move is applied to the first-principles plant and the procedure is repeated. The baseline is a PI controller on current with fuel fed at a fixed utilisation of 0.75.

4. Results

Operating envelope. Fig. 1 shows the steady map. At $U_f = 0.5$ the maximum safe power is 45.2 W, but at $U_f = 0.8$ it falls to 33.2 W because the TPB H_2 fraction reaches its limit (white dashed line) before the fuel supply is exhausted. To deliver high power safely, a controller must lower the utilisation.

Surrogate accuracy. On unseen test trajectories the GRU reaches $R^2 \geq 0.9998$ for x_{ch} , T and x_{TPB} , and remains accurate

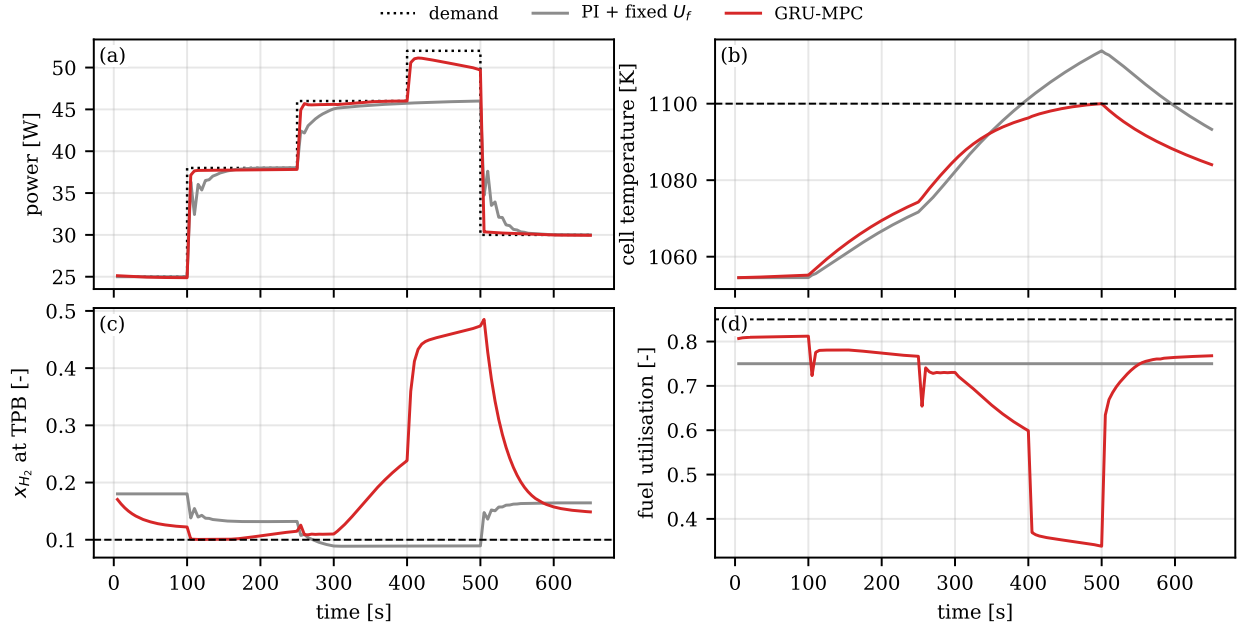


Figure 2: Closed loop on the first-principles plant for a demand of 25→38→46→52→30 W. Dashed lines are the constraints. The 52 W step lies outside the safe envelope.

over 80-step open-loop rollouts ($R^2 \geq 0.998$) although trained on 20-step windows. The voltage error is 3.6 mV RMS in the operating region; larger errors occur only in the 2.8 % of samples with a starved TPB, which the controller is designed to avoid.

Closed loop. Fig. 2 and Table 1 compare the controllers over 650 s. The baseline keeps U_f fixed, starves the TPB for 230 s and overheats the cell to 1114 K. The MPC anticipates the TPB limit, lowers utilisation before the 52 W step and holds the temperature at its bound, delivering the power that remains feasible. The price is higher fuel consumption: the baseline’s higher efficiency is obtained only by violating both limits.

Table 1: Closed-loop performance (650 s).

	PI + fixed U_f	GRU-MPC
Tracking RMSE [W]	2.75	0.64
Max. temperature [K]	1113.8	1100.0
Min. x_{TPB} [-]	0.089	0.101
Time below x_{TPB} limit [s]	230	0
Electrical efficiency (LHV) [%]	34.6	29.6

Computational cost. Over the first demand step (80–140 s), the same MPC using the first-principles model as predictor gives nearly identical inputs (mean current difference 45 A m⁻²) but needs 85 s per step, against 0.64 s for the GRU-MPC (about 130× faster on the same machine). Since the first-principles predictor uses finite-difference gradients, this figure reflects both the cheaper model and exact gradients.

5. Conclusions and outlook

A GRU surrogate trained on DGM-based simulations allows an MPC to respect a porous-transport constraint that a conventional controller violates, at a cost compatible with real-time use. The study is limited: the plant and the training data come from the same model, all states are assumed measured, parameters are literature-typical rather than fitted to a cell,

and no formal closed-loop stability guarantee is given. Natural next steps are plant–model mismatch with online model updating, state estimation from voltage and temperature, and physics-informed training.

Code availability. Code, trained model and scripts reproducing all results: github.com/ramakrishnan108/sofc-rnn-mpc. The implementation was developed with AI-assisted coding.

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